

Numerical Physics — Exercise 3
Gravitational systems with one, two and three bodies:
Artemis II trajectory and atmospheric re-entry

Pasqual Hernández Riveiro Jordi Mitjà Cortés
`pasqual.hernandezriveiro@epfl.ch` `jordi.mitjacortes@epfl.ch`

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1 Introduction

In this work, we study gravitational systems of one, two and three bodies in two spatial dimensions, with a particular focus on the Artemis II return trajectory toward the Earth. The physical model is based on Newtonian gravitation and is later extended by including the aerodynamic drag produced by the terrestrial atmosphere. The system is integrated with a fourth-order Runge–Kutta scheme in Cartesian coordinates, together with an adaptive time-step method.

The report is organised as follows. Section 2 presents the numerical formulation of the problem and the main implementation choices used in the C++ code. Section 3 is devoted to the simulations and analysis of the different physical situations proposed in the exercise. For the Artemis return problem, we first determine analytically the initial velocity required for re-entry and then compare this reference solution with fixed-step and adaptive numerical simulations. The later sections address atmospheric effects, the Earth–Moon system, and the full three-body Earth–Moon–Artemis configuration.

2 Numerical method and implementation

2.1 Gravitational interaction and atmospheric drag

We restrict ourselves to the two-dimensional case and use Cartesian coordinates throughout. The N -body problem, with $N \leq 3$, is written as a first-order system of ordinary differential equations,

$$\frac{dy}{dt} = f(y).$$

For each body i , with position (x_i, y_i) and velocity $(v_{x,i}, v_{y,i})$, the dynamical variables are

$$(x_i, y_i, v_{x,i}, v_{y,i}).$$

In addition, the gravitational force exerted on body i by body j is given by

$$\vec{F}_{ij} = -\frac{Gm_i m_j}{|\vec{r}_j - \vec{r}_i|^3} (\vec{r}_i - \vec{r}_j),$$

where G is the gravitational constant and $\vec{r}_i = (x_i, y_i)$. Dividing by m_i , the corresponding contribution to the acceleration of body i is

$$\vec{a}_{ij} = Gm_j \frac{\vec{r}_j - \vec{r}_i}{|\vec{r}_j - \vec{r}_i|^3}.$$

The total gravitational acceleration of body i is therefore obtained by summing over all other bodies:

$$\vec{a}_i^{\text{grav}} = \sum_{j \neq i} Gm_j \frac{\vec{r}_j - \vec{r}_i}{|\vec{r}_j - \vec{r}_i|^3}.$$

In the sections devoted to atmospheric re-entry, the model is extended by adding the drag force

$$\vec{F}_d = -\frac{1}{2} \rho(r) S C_x |\vec{v}_i - \vec{v}_j| (\vec{v}_i - \vec{v}_j),$$

where S is the cross-sectional area of the spacecraft, C_x the drag coefficient, and $\rho(r)$ the local atmospheric density. The latter is modeled by the exponential law

$$\rho(r) = \rho_0 \exp\left(-\frac{r - R_T}{\lambda}\right),$$

with ρ_0 the sea-level density, R_T the Earth radius and λ the characteristic atmospheric scale height. In practice, this drag term is applied only to the spacecraft and only when the atmosphere is activated in the corresponding simulations.

The acceleration used in the numerical integration is thus the sum of the gravitational contribution and, when applicable, the drag contribution:

$$\vec{a}_i = \vec{a}_i^{\text{grav}} + \vec{a}_i^{\text{drag}}.$$

2.2 Fourth-order Runge–Kutta scheme

Time integration is performed with the classical fourth-order Runge–Kutta method. Starting from the state y_n at time t_n , the value at $t_{n+1} = t_n + \Delta t$ is computed as

$$y_{n+1} = y_n + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4),$$

with

$$\begin{aligned} k_1 &= f(y_n), \\ k_2 &= f\left(y_n + \frac{\Delta t}{2} k_1\right), \\ k_3 &= f\left(y_n + \frac{\Delta t}{2} k_2\right), \\ k_4 &= f(y_n + \Delta t k_3). \end{aligned}$$

This method is particularly appropriate here because the equations are already written as a first-order system and the force model can be evaluated explicitly once the state vector is known. In the code, the four k -vectors are stored in auxiliary arrays, and each RK4 step is obtained through successive evaluations of the derivative function $f(y)$.

2.3 Adaptive time stepping

Besides fixed-step simulations, we also implement an adaptive time-step scheme. The idea is to reduce Δt in regions where the trajectory varies rapidly, for example near the Earth, and to allow larger values when the motion is smoother.

The procedure compares two approximations of the solution over the same physical interval: one obtained with a single step Δt , and one obtained with two successive half-steps $\Delta t/2$. The difference between these two estimates provides an error indicator d . If $d < \varepsilon$, where ε is the required tolerance, the step is accepted; otherwise it is rejected and recomputed with a smaller value of Δt .

According to the course notes, the new time step is updated through a rescaling law of the form

$$\Delta t_{\text{new}} = s \Delta t \left(\frac{\varepsilon}{d} \right)^{1/(n+1)},$$

where $n = 4$ for RK4 and $s < 1$ is a safety factor. This approach improves efficiency while maintaining the required accuracy along the trajectory.

2.4 Mechanical energy and total momentum

As requested in the exercise, the code also computes the mechanical energy and the total linear momentum of the system.

The total mechanical energy is written as the sum of the kinetic energy of all bodies and the gravitational potential energy of all interacting pairs:

$$E_{\text{mec}} = \sum_{i=1}^N \frac{1}{2} m_i (v_{x,i}^2 + v_{y,i}^2) - \sum_{i < j} \frac{G m_i m_j}{|\vec{r}_i - \vec{r}_j|}.$$

In the conservative cases, this quantity should remain constant up to numerical errors and is therefore a useful diagnostic for validating the implementation.

The total linear momentum is

$$\vec{P} = \sum_{i=1}^N m_i \vec{v}_i.$$

For isolated gravitational systems, it should also be conserved. Monitoring E_{mec} and $\vec{P}$ is particularly useful in the two-body Earth–Moon problem, where these conservation properties provide a direct check of the numerical solver.

2.5 Structure of the code and outputs

Following the programming guide, the numerical tool is organised around three main stages: initialization of the physical and numerical parameters, temporal evolution of the system, and post-processing of the output data. The constructor reads the parameters from a configuration file using the same `ConfigFile` logic as in the previous exercises. The time evolution is handled by the method `run()`, which repeatedly calls the time-stepping routine and writes the results to an output file at a chosen sampling frequency. Depending on the control parameter `timeScheme`, the code uses either a fixed time step or the adaptive strategy described above. At each sampled instant, the program stores the current time, the positions and velocities of all bodies, and the derived quantities such as mechanical energy, total momentum, and, when relevant, the power associated with aerodynamic drag. These files are then analysed in Python in order to extract the relevant observables: trajectories, minimum altitude, maximum speed, convergence curves, time-step evolution, and conservation diagnostics.

3 Simulations and analyses

3.1 Artemis II return trajectory

3.1.1 (a) Analytical determination of the initial velocity and maximum speed

In this first part, we neglect both the Moon and the atmosphere, so Artemis II is treated as a test particle moving only in the gravitational field of the Earth. The goal is to determine the initial velocity direction such that the resulting orbit has a minimum altitude

$$h = 10000 \text{ m}$$

above the Earth’s surface.

We denote by

$$\mu = G m_T$$

the Earth’s gravitational parameter, with

$$m_T = 5.972 \times 10^{24} \text{ kg}, \quad G = 6.6740 \times 10^{-11} \text{ SI},$$

which gives

$$\mu = 3.9857 \times 10^{14} \text{ m}^3/\text{s}^2.$$

The initial distance from the Earth's centre is

$$r_0 = 314159 \text{ km} = 3.14159 \times 10^8 \text{ m},$$

and the initial speed magnitude is fixed to

$$|\vec{v}_0| = 1.2 \text{ km/s} = 1200 \text{ m/s}.$$

The required periapsis distance is therefore

$$r_p = R_T + h = 6.3881 \times 10^6 \text{ m}.$$

The specific mechanical energy of the orbit is

$$\varepsilon = \frac{v_0^2}{2} - \frac{\mu}{r_0} \approx -5.4869 \times 10^5 \text{ J/kg}.$$

Since $\varepsilon < 0$, the orbit is elliptical, and the semi-major axis is

$$a = -\frac{\mu}{2\varepsilon} \approx 3.6320 \times 10^8 \text{ m}.$$

Using

$$r_p = a(1 - e),$$

we obtain the eccentricity

$$e = 1 - \frac{r_p}{a} \approx 0.9824.$$

The specific angular momentum is then

$$h_{\text{orb}} = \sqrt{\mu a(1 - e^2)} \approx 7.1045 \times 10^{10} \text{ m}^2/\text{s}.$$

At the initial point, the tangential velocity component satisfies

$$h_{\text{orb}} = r_0 v_{t0},$$

hence

$$v_{t0} = \frac{h_{\text{orb}}}{r_0} \approx 226.15 \text{ m/s}.$$

Since the total initial speed is fixed, the radial component is

$$v_{r0} = \sqrt{v_0^2 - v_{t0}^2} \approx 1178.50 \text{ m/s}.$$

Because Artemis must initially move toward the Earth, this component must be negative:

$$v_{r0} \approx -1178.50 \text{ m/s}.$$

Choosing the initial position on the positive x -axis,

$$\vec{r}_0 = (r_0, 0),$$

the radial direction coincides with the x -axis and the tangential direction with the y -axis. A convenient choice for the initial velocity is therefore

$$\boxed{v_{x0} = -1178.50 \text{ m/s}, \quad v_{y0} = 226.15 \text{ m/s}.}$$

The symmetric choice with $v_{y0} < 0$ would produce the same orbit reflected with respect to the x -axis.

Finally, the maximum speed is reached at periapsis, where the velocity is purely tangential:

$$v_{\text{max}} = \frac{h_{\text{orb}}}{r_p}.$$

This gives

$$\boxed{v_{\text{max}} \approx 1.112 \times 10^4 \text{ m/s} = 11.12 \text{ km/s}.}$$

These analytical values provide the reference solution used in the next subsections to validate the fixed-step and adaptive numerical simulations.

3.1.2 (b) Fixed-step RK4 simulation and convergence study

We now solve the Artemis II trajectory numerically using the fixed-step RK4 scheme described in Section 2. In this part, the atmosphere is still neglected and the Moon is taken with a negligible mass, so that the motion remains essentially that of a spacecraft evolving in the gravitational field of the Earth alone. The numerical results are therefore expected to converge toward the analytical solution obtained in the previous subsection.

The simulations were performed over a total duration of

$$t_{\text{end}} = 2 \text{ days} = 172800 \text{ s},$$

which is sufficient to capture the approach to the Earth and the passage through periapsis. The initial position of Artemis was taken as

$$(x_0, y_0) = (3.14159 \times 10^8 \text{ m}, 0),$$

and the initial velocity components were set to the analytical values found in part (a),

$$v_{x0} = -1178.50 \text{ m/s}, \quad v_{y0} = 226.15 \text{ m/s}.$$

A series of simulations was then carried out with different fixed time steps Δt , ranging from coarse to fine resolution, in order to study the convergence of the numerical results.

Trajectory. Figure 1 shows the Artemis trajectory in the (x, y) plane for three representative values of the time step. On the scale of the figure, the three curves are indistinguishable, which indicates that the numerical solution is stable and that the overall orbital shape is well reproduced even for moderately large values of Δt . A more quantitative assessment of the convergence is obtained from the errors on the minimum altitude and on the maximum speed, discussed below.

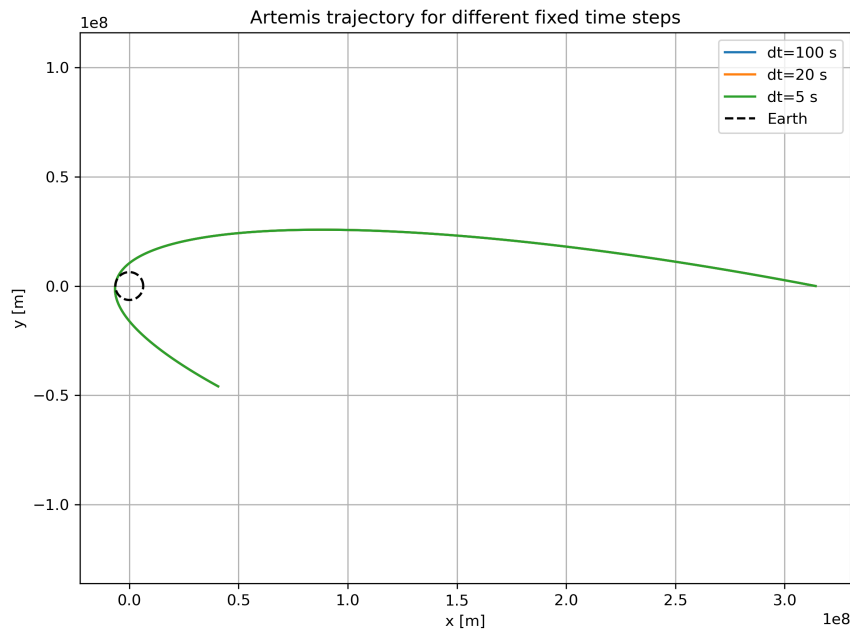


Figure 1: Artemis trajectory in the (x, y) plane for three representative fixed time steps.

Minimum altitude. To quantify the convergence, we extracted from each simulation the minimum altitude

$$h_{\min} = \min_t (r(t) - R_T),$$

where

$$r(t) = \sqrt{(x_A - x_T)^2 + (y_A - y_T)^2}$$

is the instantaneous distance between Artemis and the Earth. The analytical target value is

$$h_{\min, \text{ana}} = 10000 \text{ m.}$$

The error

$$\varepsilon_h = |h_{\min, \text{num}} - h_{\min, \text{ana}}|$$

is shown in Figure 2. The error decreases clearly as Δt is reduced, which confirms that the numerical trajectory converges toward the analytical re-entry orbit. For large time steps, the periapsis is not resolved accurately enough, which leads to sizeable deviations in the estimated minimum altitude. In contrast, once Δt becomes sufficiently small, the numerical value of $h_{\min}$ approaches the analytical one.

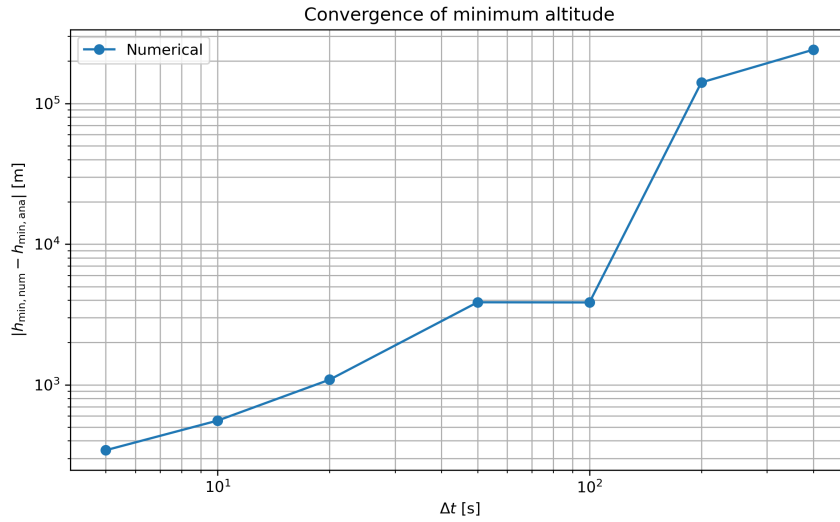


Figure 2: Absolute error on the minimum altitude as a function of the fixed time step Δt .

Maximum speed. A second diagnostic is the maximum spacecraft speed,

$$v_{\max} = \max_t \sqrt{v_x^2 + v_y^2},$$

which should converge toward the analytical value obtained in part (a),

$$v_{\max, \text{ana}} \approx 11.12 \text{ km/s.}$$

The corresponding error,

$$\varepsilon_v = |v_{\max, \text{num}} - v_{\max, \text{ana}}|,$$

is plotted in Figure 3. Here again, the numerical error decreases as the time step is refined. This confirms that the fixed-step RK4 scheme captures not only the global shape of the orbit, but also the correct dynamical behaviour near periapsis, where the speed reaches its maximum value.

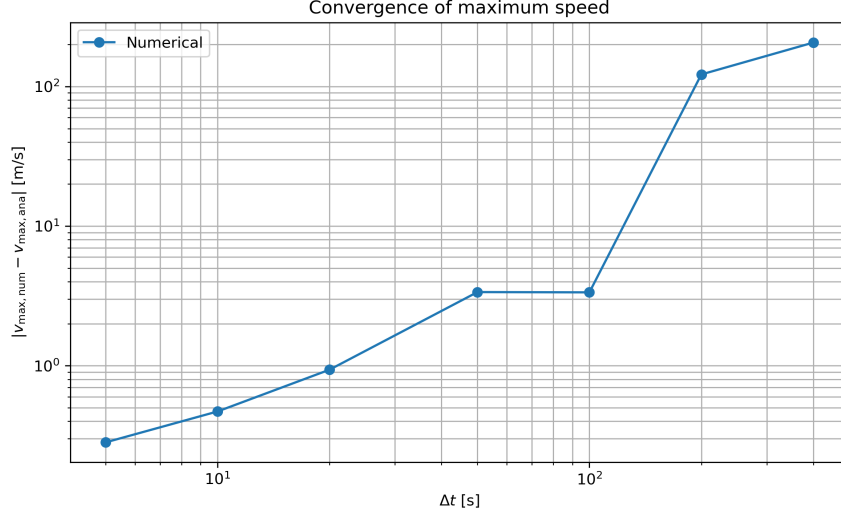


Figure 3: Absolute error on the maximum speed as a function of the fixed time step Δt .

Overall, these results validate the implementation of the fixed-step RK4 solver in the simplest Artemis return configuration. The trajectory is reproduced correctly, and both $h_{\min}$ and $v_{\max}$ converge toward their analytical reference values as the discretisation is refined.

3.1.3 (c) Adaptive RK4 simulation and timestep evolution

We now study the same Artemis return problem using the adaptive RK4 scheme. The physical setting is unchanged: the atmosphere is neglected and the Moon is given a negligible mass, so that the trajectory remains essentially determined by the Earth's gravitational field. The difference with the previous subsection is purely numerical: instead of fixing the time step Δt , we let the code adjust it automatically according to a prescribed tolerance ε .

A series of simulations was performed for different values of ε , while keeping the same initial conditions as in parts (a) and (b). For each run, we extracted the minimum altitude

$$h_{\min} = \min_t (r(t) - R_T)$$

and the maximum speed

$$v_{\max} = \max_t \sqrt{v_x^2 + v_y^2},$$

and compared them with the analytical reference values obtained in part (a).

Convergence with respect to the tolerance. Figure 4 shows the absolute error on the minimum altitude as a function of the tolerance ε . The error remains small over the whole range explored and decreases overall as the tolerance is made more restrictive. The same behaviour is observed for the maximum speed in Figure 5. Although the curves are not perfectly monotonic, the general trend is clear: a smaller tolerance leads to a more accurate numerical solution. The residual fluctuations are attributed to the fact that both $h_{\min}$ and $v_{\max}$ are extracted as extrema of the numerically sampled trajectory, so their determination is slightly sensitive to the exact sampling near periapsis.

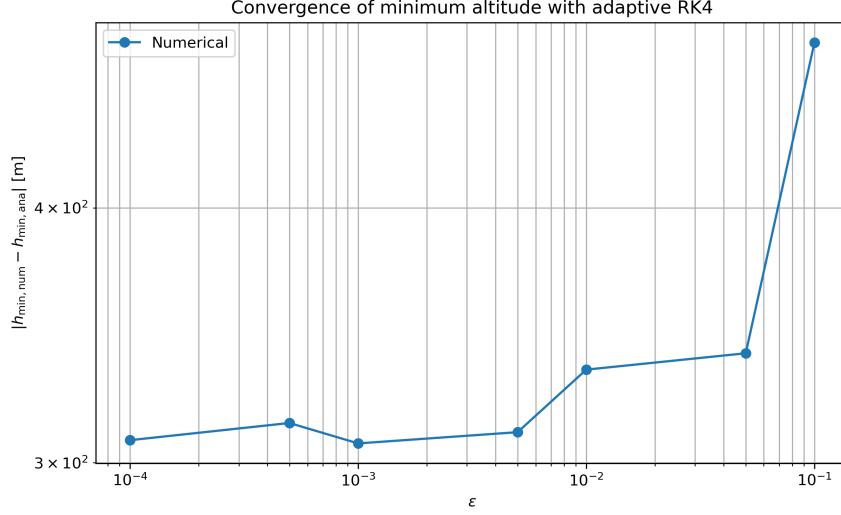


Figure 4: Absolute error on the minimum altitude as a function of the tolerance ε for the adaptive RK4 scheme.

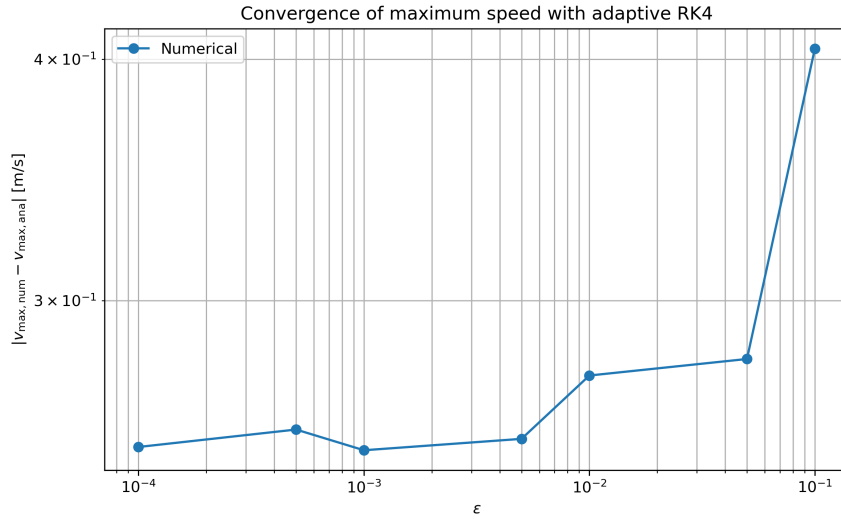


Figure 5: Absolute error on the maximum speed as a function of the tolerance ε for the adaptive RK4 scheme.

Number of time steps. The main advantage of the adaptive scheme is illustrated in Figure 6, where we plot the number of time steps used during the simulation as a function of ε . As expected, the number of steps increases when the tolerance is reduced. This confirms that the adaptive method automatically invests more computational effort when a higher accuracy is required, while using fewer steps when a looser tolerance is sufficient.

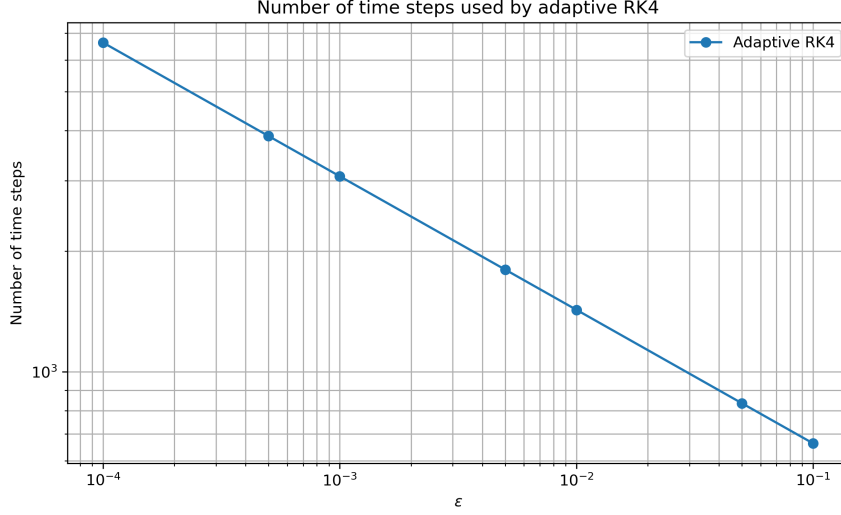


Figure 6: Number of time steps used by the adaptive RK4 method as a function of the tolerance ε .

Time-step evolution along the trajectory. To understand how the adaptive method distributes its effort, we analyse a representative simulation with

$$\varepsilon = 10^{-3}.$$

The time evolution of the adaptive step is shown in Figure 7. Starting from the initial value, the time step first increases while Artemis remains far from the Earth, where the dynamics is smooth and slowly varying. As the spacecraft approaches periapsis, Δt decreases significantly, reaching its smallest values near the point of closest approach. After periapsis, the step increases again as the spacecraft moves away and the trajectory becomes less curved.

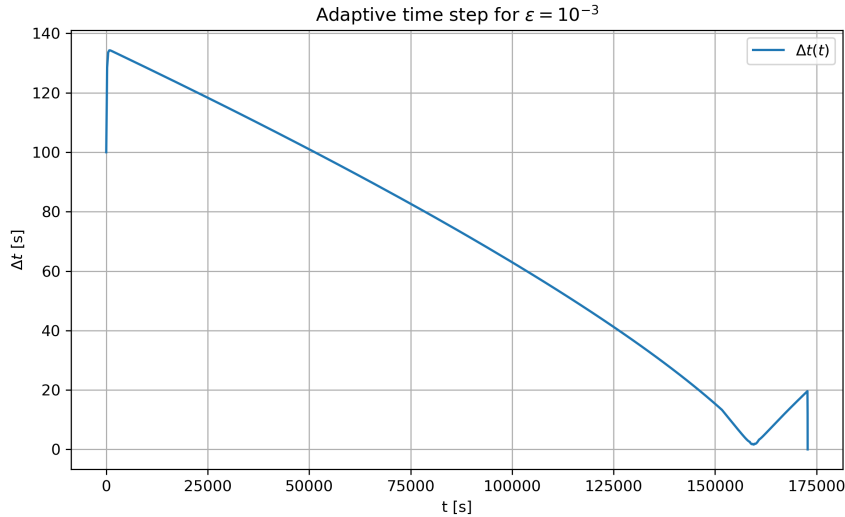


Figure 7: Time evolution of the adaptive time step for the representative run with $\varepsilon = 10^{-3}$.

This behaviour is even more clearly visible in Figure 8, where Δt is plotted directly as a function of the distance to the Earth. The numerical method uses large time steps when Artemis is far away and much smaller ones when the spacecraft is close to the Earth. This is precisely the expected behaviour of an adaptive integrator: it concentrates the numerical effort in the most

demanding part of the orbit, namely the region near periapsis where the acceleration varies most rapidly.

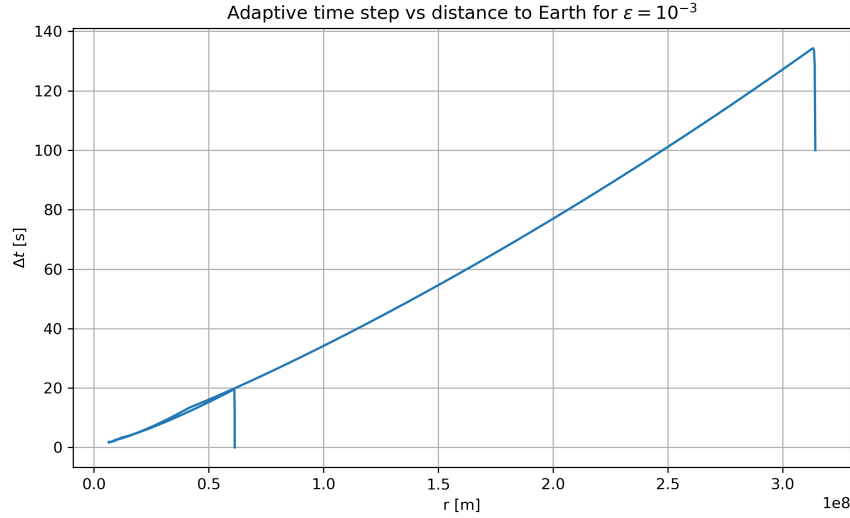


Figure 8: Adaptive time step as a function of the distance to the Earth for the representative run with $\varepsilon = 10^{-3}$.

Comparison with the fixed-step scheme. The results obtained here are consistent with those of part (b). Both methods converge toward the analytical solution, but the adaptive method has the advantage of adjusting the resolution automatically along the trajectory. In particular, it avoids using an unnecessarily small time step during the whole simulation, while still refining the integration close to the Earth. This makes it more efficient than the fixed-step method for a highly non-uniform trajectory such as the present re-entry orbit.

Overall, the adaptive RK4 scheme reproduces the analytical reference values of $h_{\min}$ and $v_{\max}$ with good accuracy, while reducing the computational cost by adapting the time step to the local difficulty of the trajectory.

3.2 Atmospheric re-entry

3.2.1 (a) Re-entry with atmospheric drag

We now consider the same Artemis return trajectory as in Section 3.1, but including the effect of atmospheric drag. The Moon is still given a negligible mass, so that the motion is essentially governed by the Earth's gravity and by the dissipative force due to the atmosphere. The initial conditions are kept identical to those determined in Section 3.1(a), namely

$$(x_0, y_0) = (3.14159 \times 10^8 \text{ m}, 0),$$

$$v_{x0} = -1178.50 \text{ m/s}, \quad v_{y0} = 226.15 \text{ m/s},$$

and the simulations are performed with the adaptive RK4 scheme.

The purpose of this subsection is to study how the inclusion of the atmosphere affects the re-entry and, in particular, to analyse two quantities of practical importance: the maximum acceleration experienced by Artemis and the maximum power dissipated by the drag force.

Trajectory and adaptive time step. Figure 9 shows the resulting trajectory in the (x, y) plane for a representative run with

$$\varepsilon = 10^{-3}.$$

On the scale of the full orbit, the trajectory remains globally similar to the one obtained without atmosphere, which is expected since the atmospheric force acts only very close to the Earth. The drag therefore has a negligible influence over most of the path, but becomes important during the final part of the approach.

This localisation is also reflected in the adaptive time step shown in Figure 10. Far from the Earth, the motion is smooth and the integrator uses relatively large time steps. As Artemis approaches the dense layers of the atmosphere, the step decreases dramatically, reaching very small values in the short interval where the re-entry dynamics becomes most demanding. This is precisely the expected behaviour of the adaptive scheme: it concentrates the numerical effort in the region where both gravity and drag vary most rapidly.

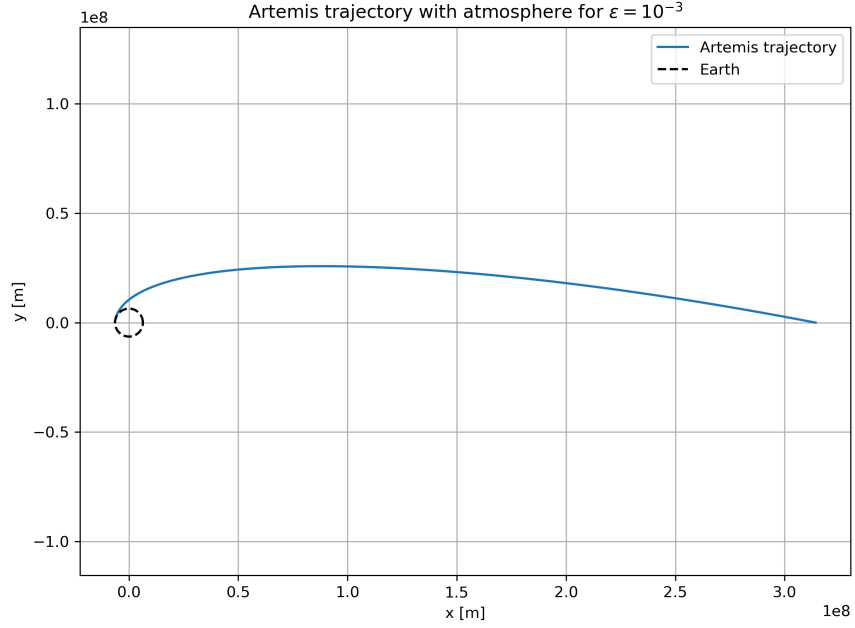


Figure 9: Artemis trajectory with atmospheric drag for a representative simulation with $\varepsilon = 10^{-3}$.

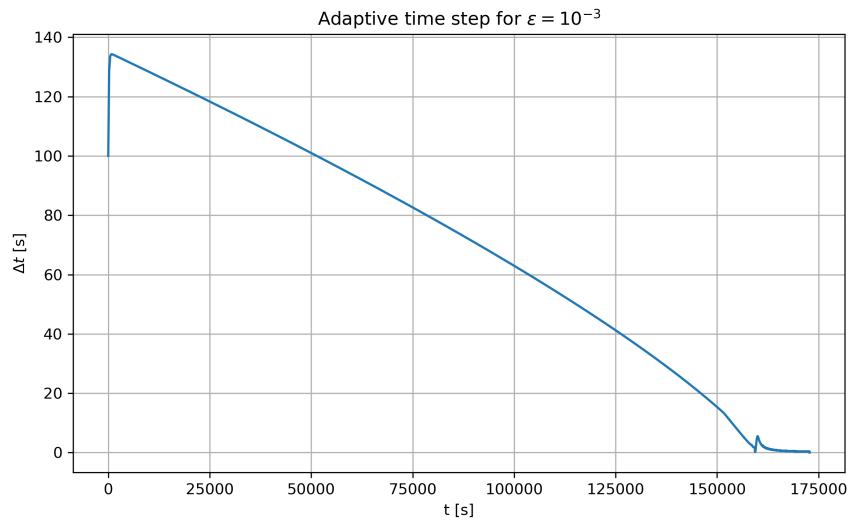


Figure 10: Adaptive time step as a function of time for the representative simulation with $\varepsilon = 10^{-3}$.

Instantaneous acceleration and drag power. The time evolution of the instantaneous acceleration is shown in Figure 11. For most of the trajectory, the acceleration remains modest on the scale of the plot, but a very sharp peak appears near the re-entry stage. This indicates that the strongest mechanical stress is concentrated in a very short time interval close to the closest approach to the Earth.

A similar behaviour is observed for the drag power in Figure 12. The magnitude of the dissipated power is essentially negligible during most of the trajectory and becomes extremely large only in a narrow temporal window around re-entry. This is physically reasonable: the drag force depends on both the local atmospheric density and the relative speed, so the strongest dissipation occurs only when Artemis simultaneously reaches dense atmospheric layers and still has a very large velocity.

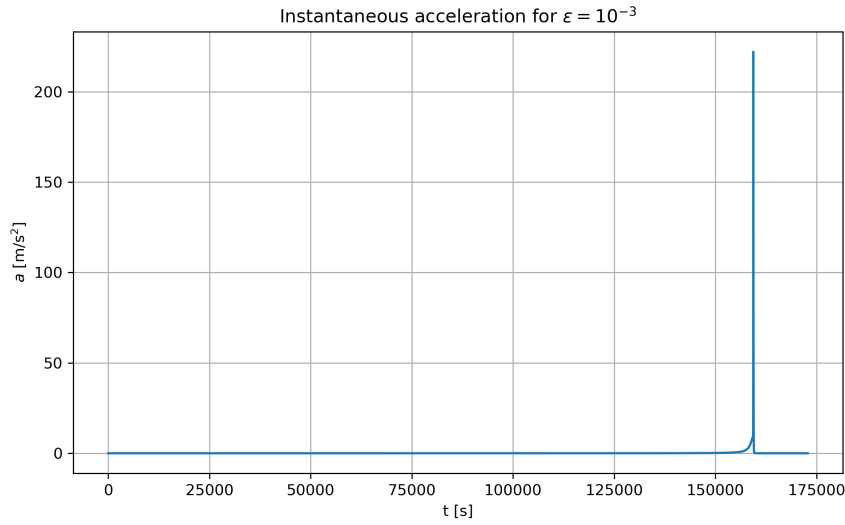


Figure 11: Instantaneous acceleration of Artemis as a function of time for the representative simulation with $\varepsilon = 10^{-3}$.

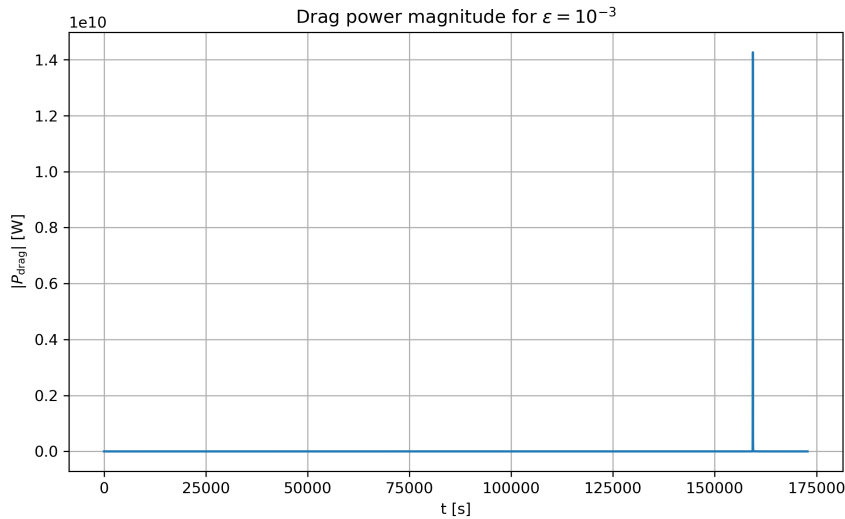


Figure 12: Magnitude of the drag power as a function of time for the representative simulation with $\varepsilon = 10^{-3}$.

Convergence of the maximum acceleration and maximum drag power. To check that these observables are numerically well resolved, we repeated the adaptive simulation for several values of the tolerance ε . From each run, we extracted the maximum acceleration

$$a_{\max} = \max_t a(t)$$

and the maximum magnitude of the drag power

$$|P_{\text{drag}}|_{\max} = \max_t |P_{\text{drag}}(t)|.$$

The results are shown in Figures 13 and 14. In both cases, the variation with the tolerance is small over the explored range, which indicates that the two quantities are already well converged. The maximum acceleration remains close to the same value for all tolerances considered, and the same is true for the maximum drag power. This shows that the adaptive RK4 scheme captures the short and intense re-entry event in a robust way.

Overall, the inclusion of the atmosphere leaves the large-scale trajectory almost unchanged, but introduces a highly localised dissipative phase characterised by a strong peak in acceleration and an even more concentrated peak in drag power. The adaptive integrator is particularly well suited to this situation, since it automatically refines the time step exactly where the dynamics becomes stiffest.

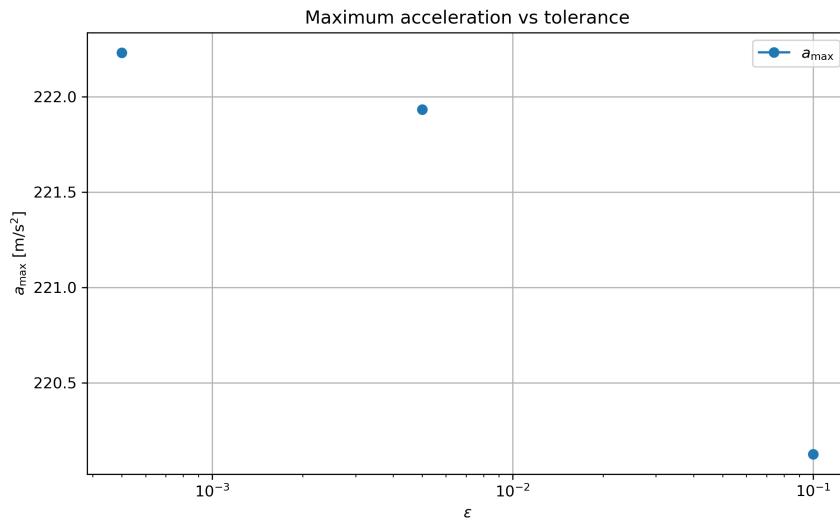


Figure 13: Maximum acceleration as a function of the adaptive tolerance ε .

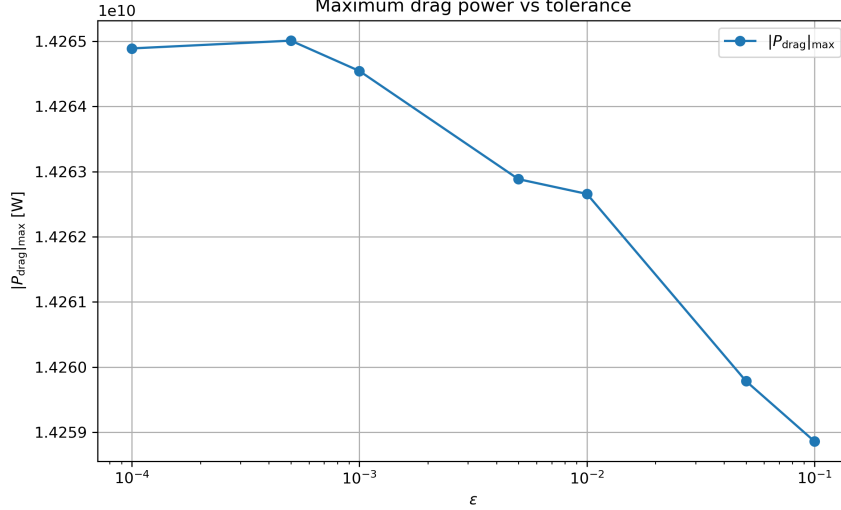


Figure 14: Maximum magnitude of the drag power as a function of the adaptive tolerance ε .

3.2.2 (b) Optimisation of the initial direction

In this subsection, we keep the same physical model as in part (a): the atmosphere is included, the Moon is still given a negligible mass, and the Artemis trajectory is integrated with the adaptive RK4 scheme. The objective is now to vary the initial direction of the spacecraft velocity in order to reduce the maximal acceleration experienced during re-entry.

The initial position is kept fixed at

$$(x_0, y_0) = (3.14159 \times 10^8 \text{ m}, 0),$$

and the magnitude of the initial velocity is also kept fixed at

$$|\vec{v}_0| = 1200 \text{ m/s}.$$

Only its direction is varied. More precisely, we introduce an angular shift $\Delta\theta$ with respect to the reference direction found in Section 3.2(a), and for each value of $\Delta\theta$ we compute the corresponding trajectory, its minimum altitude $h_{\min}$, its maximum acceleration $a_{\max}$, and the maximum magnitude of the drag power $|P_{\text{drag}}|_{\max}$.

Criterion for a valid re-entry. A first important point is that not all angular shifts correspond to a meaningful atmospheric re-entry. For sufficiently negative values of $\Delta\theta$, Artemis does not penetrate deeply enough into the atmosphere, and the resulting trajectory no longer represents the same type of re-entry problem. In order to restrict the study to physically relevant cases, we classify as valid those trajectories whose minimum altitude satisfies

$$h_{\min} < h_{\text{reentry},\max}, \quad h_{\text{reentry},\max} = 120 \text{ km}.$$

Figure 15 shows the dependence of the minimum altitude on the initial angular shift. Over the scanned interval, $h_{\min}$ decreases as $\Delta\theta$ increases: for more negative angular shifts, the trajectory passes farther from the Earth and the minimum altitude is larger, whereas angles close to the reference value and more positive shifts lead to much deeper atmospheric passages. This explains why sufficiently negative angles correspond to invalid trajectories, while angles near the reference direction and above it remain within the selected re-entry regime.

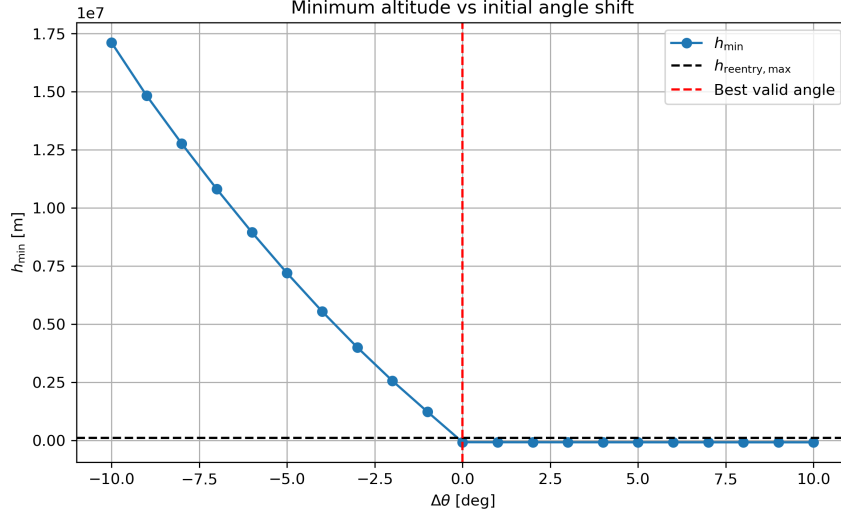


Figure 15: Minimum altitude as a function of the initial angular shift $\Delta\theta$. The dashed horizontal line indicates the threshold used to define a valid re-entry trajectory.

Maximum acceleration. We then restrict the optimisation to the valid trajectories and study the corresponding maximum acceleration. The result is shown in Figure 16. Among the valid re-entry trajectories, the minimum value of $a_{\max}$ is obtained for an angular shift very close to the reference one,

$$\Delta\theta \approx 0^\circ.$$

This means that the initial direction derived in Section 3.2(a) is already essentially optimal with respect to the criterion of minimizing the peak acceleration, provided that one still requires a genuine atmospheric re-entry.

For positive angular shifts, the trajectory penetrates more deeply into the atmosphere and the maximal acceleration increases rapidly. On the other hand, the very small values of $a_{\max}$ observed for some negative shifts are not physically relevant for the optimisation, since those trajectories are precisely the ones that fail the re-entry criterion discussed above.

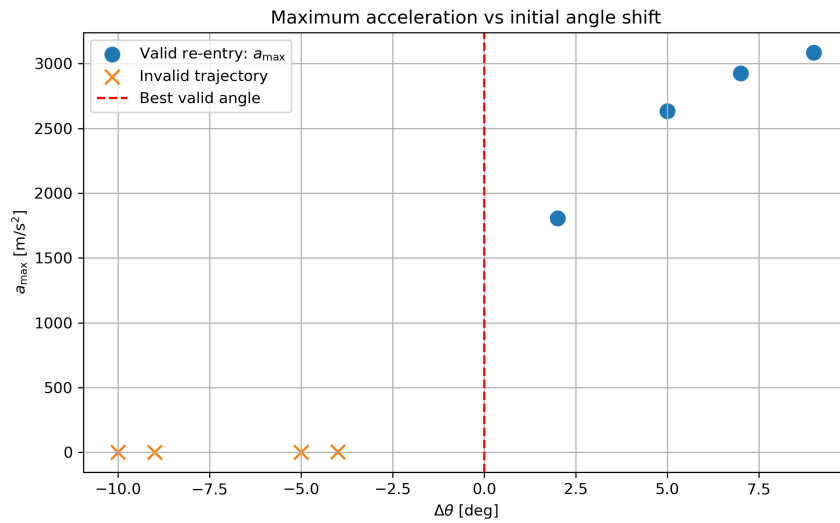


Figure 16: Maximum acceleration as a function of the initial angular shift $\Delta\theta$. Valid and invalid trajectories are distinguished explicitly. The dashed vertical line marks the best valid angle.

Maximum drag power. The same trend is observed for the maximum magnitude of the drag power, shown in Figure 17. For invalid trajectories, the drag remains negligible because Artemis does not reach sufficiently dense atmospheric layers. In contrast, once the trajectory is deep enough to qualify as a valid re-entry, the drag power becomes non-zero and then grows strongly with increasing $\Delta\theta$. This again reflects the fact that more positive angles correspond to stronger atmospheric penetration and therefore to more intense dissipation.

The consistency between Figures 16 and 17 is physically expected: a deeper atmospheric passage simultaneously increases the dissipative power and the peak deceleration experienced by the spacecraft.

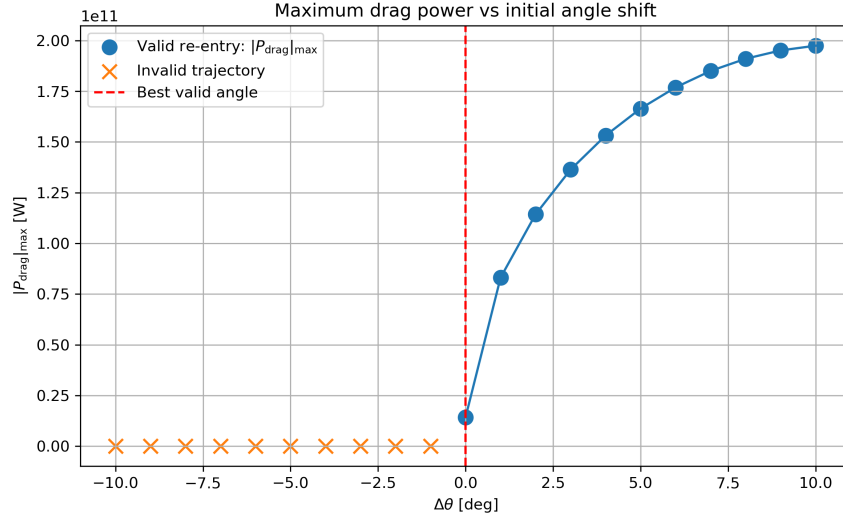


Figure 17: Maximum magnitude of the drag power as a function of the initial angular shift $\Delta\theta$. Valid and invalid trajectories are distinguished explicitly.

Selected optimal trajectory. Since the minimum of a_{max} over the valid set is obtained at the reference angle, the optimal trajectory is essentially the same as the one used in part (a). Its overall shape is shown in Figure 18. On the scale of the full orbit, it remains close to the previously studied return trajectory, while the decisive differences occur only during the short atmospheric interaction near the Earth.

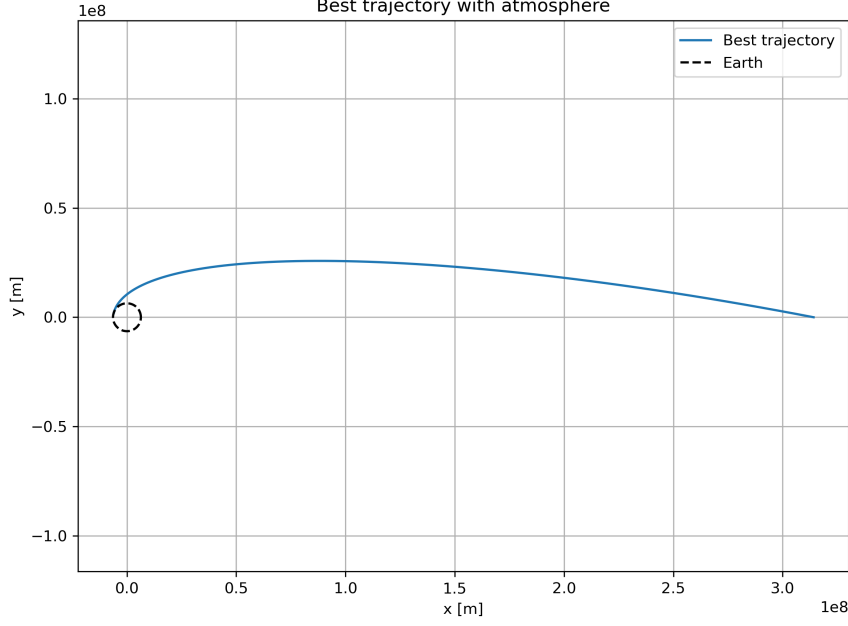


Figure 18: Trajectory corresponding to the optimal valid initial direction, i.e. the one minimizing the maximal acceleration among the atmospheric re-entry trajectories.

Overall, this optimisation study shows that decreasing the peak acceleration is only possible up to the point where the trajectory becomes too shallow to remain in the targeted atmospheric re-entry regime. Once the validity condition is imposed, the reference initial direction found previously remains essentially optimal. More deeply penetrating trajectories lead to significantly larger acceleration and drag power, while shallower ones eventually cease to represent the intended re-entry scenario.

3.3 Earth–Moon system

3.3.1 (a) Analytical initial conditions for an Earth–Moon system

We now consider the Earth–Moon system with their real masses, neglecting the atmosphere. We need to find the initial conditions, position and velocity for the Moon and the Earth, corresponding to a uniform circular trajectory in the center of mass frame, with constant Earth–Moon distance

$$d = 384748 \text{ km.}$$

Since we are working in the frame of the center of mass, we place the origin at the center of mass of the system. First of all, we need to find where this center is with respect to the Earth and the Moon.

For a circular motion in the center-of-mass frame, the Earth and the Moon must remain on opposite sides of the origin, at distances r_T and r_L , such that they keep the same distance at all times

$$r_T + r_L = d. \quad (1)$$

The equation to find the center of mass is

$$\vec{R}_{CM} = \frac{m_T \vec{r}_T + m_L \vec{r}_L}{m_T + m_L}$$

However, since the center of mass is at the origin, we end up with

$$m_T r_T = m_L r_L. \quad (2)$$

Where m_T and m_L denote the masses of the Earth and the Moon, respectively. Combining (1) and (2) conditions, we find that the distances of each body to the center of mass of the system are

$$\boxed{r_T = \frac{m_L}{m_T + m_L} d = 4676.24 \text{ km}} \quad (3)$$

$$\boxed{r_L = \frac{m_T}{m_T + m_L} d = 380071.76 \text{ km}} \quad (4)$$

We now need to determine the velocity of the motion. Starting with the gravitational force between the two bodies which is

$$F = \frac{Gm_T m_L}{d^2},$$

Indeed, this force provides the centripetal force required for the circular motion of each body. Therefore, for the Earth, we have

$$m_T \omega^2 r_T = \frac{Gm_T m_L}{d^2}.$$

Where substituting (3) found before,

$$m_T \omega^2 \left(\frac{m_L}{m_T + m_L} d \right) = \frac{Gm_T m_L}{d^2}.$$

After simplification, we obtain

$$\omega = \sqrt{\frac{G(m_T + m_L)}{d^3}}$$

The corresponding speeds are then

$$\boxed{v_T = \omega r_T = \sqrt{\frac{Gm_L^2}{d(m_T + m_L)}} = 12.45 \text{ m/s}} \quad (5)$$

$$\boxed{v_L = \omega r_L = \sqrt{\frac{Gm_T^2}{d(m_T + m_L)}} = 1011.6 \text{ m/s}} \quad (6)$$

To define the initial conditions, we choose the x -axis along the Earth–Moon line, with the Earth on the left and the Moon on the right. Since r_T and r_L are distances, they are positive quantities. The sign is introduced only when defining the position vectors. Thus, the initial positions are chosen as

$$\vec{r}_T(0) = (-r_T, 0), \quad \vec{r}_L(0) = (r_L, 0).$$

For a counterclockwise circular motion, the velocities must be perpendicular to the radius vectors. Therefore, the initial velocities are

$$\vec{v}_T(0) = (0, -v_T), \quad \vec{v}_L(0) = (0, v_L).$$

With this choice, the total momentum is zero:

$$m_T \vec{v}_T(0) + m_L \vec{v}_L(0) = \vec{0},$$

which is consistent with the center-of-mass frame.

3.3.2 (b) Numerical validation and conservation of physical properties

In the analytical two-body problem, the motion can be reduced to the motion of a single particle in a central gravitational field. For the specific initial conditions chosen here, the system is initialized in a circular orbit. In this case, the gravitational force provides exactly the centripetal force required for uniform circular motion. As a result, the angular velocity ω is constant and the relative distance d_{TL} remains constant over time. Therefore, analytically, $d_{TL} = d$ is expected.

The gravitational force is a conservative force, as it derives from a potential:

$$U(r) = -\frac{Gm_T m_L}{r}.$$

In the absence of non-conservative forces (such as drag), the total mechanical energy

$$E = T + U$$

is conserved. In the circular orbit case, both kinetic and potential energies remain constant in time, leading to a constant total mechanical energy. Any variation observed in the numerical simulation is therefore due to discretization errors of the integration scheme.

In addition, the system is isolated and only internal forces act between the Earth and the Moon. These forces are equal and opposite (Newton's third law), which implies that the total momentum

$$\mathbf{P} = m_T \mathbf{v}_T + m_L \mathbf{v}_L$$

is conserved in time. Moreover, by choosing initial conditions in the center-of-mass reference frame, we impose

$$\mathbf{P}(0) = 0,$$

which implies that $\mathbf{P}(t) = 0$ for all times. This explains why the total momentum should remain constant and close to zero throughout the simulation.

We performed numerical simulations of the Earth–Moon system over several orbital periods in order to verify that the main physical properties of the motion are satisfied: conservation of the Earth–Moon distance, conservation of mechanical energy, and conservation of total momentum. As shown in Fig. 19, the distance between the Earth and the Moon remains essentially constant over time, with only small oscillations. These oscillations are numerical in origin and remain very small compared to the mean distance. This is further confirmed in Fig. 20, where the absolute error on the Earth–Moon distance stays bounded and small throughout the simulation. Therefore, the condition $d_{TL} = d$ is well satisfied.

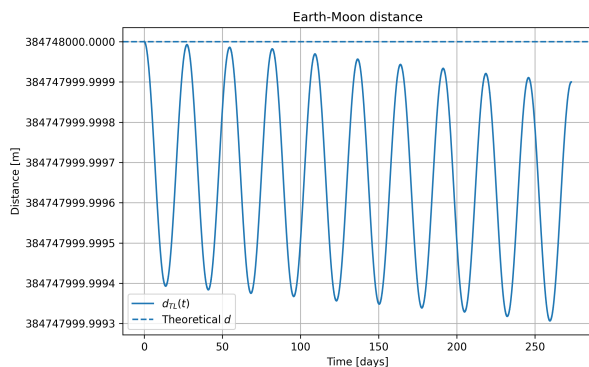


Figure 19: Distance Earth-Moon over time

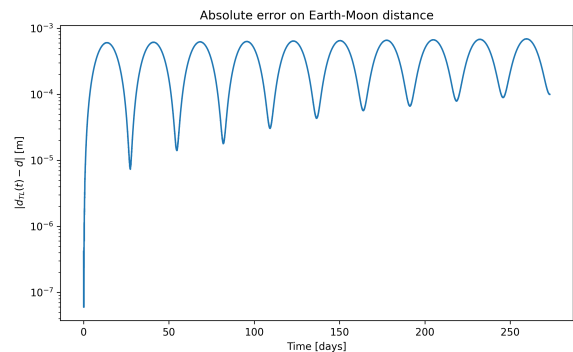


Figure 20: Absolute error on the Earth-Moon distance over time

On the other hand, Fig. 21 shows the evolution of the total mechanical energy over time. Although a slight drift is visible, it remains very small. This is better quantified in Fig. 22, where the relative error on the mechanical energy remains at the level of $\sim 10^{-13}$. This confirms that the numerical scheme preserves energy with very high accuracy, and that the observed variations are due to numerical errors associated with the finite time step of the RK4 method.

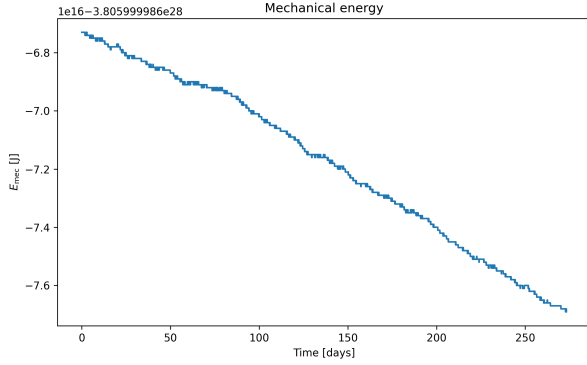


Figure 21: Mechanical energy over time

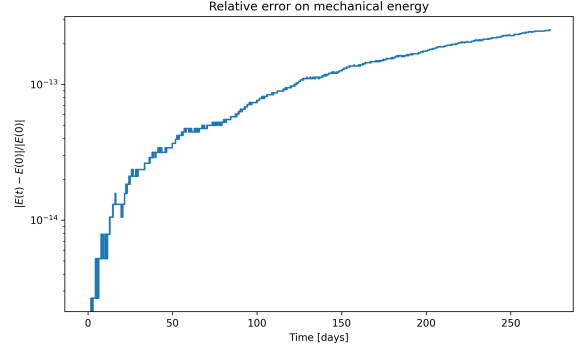


Figure 22: Relative error on the mechanical energy over time

The total momentum of the system is expected to be conserved in the absence of external forces. In the center-of-mass reference frame, its theoretical value is zero. As shown in Fig. 23, the components P_x and P_y remain close to zero, with small fluctuations of order 10^{12} kg m/s. Although this may appear large in absolute terms, it is negligible compared to the characteristic momentum scale of the system, of order 10^{25} – 10^{26} kg m/s. This corresponds to a relative level of about 10^{-13} , indicating that total momentum is very well conserved numerically.

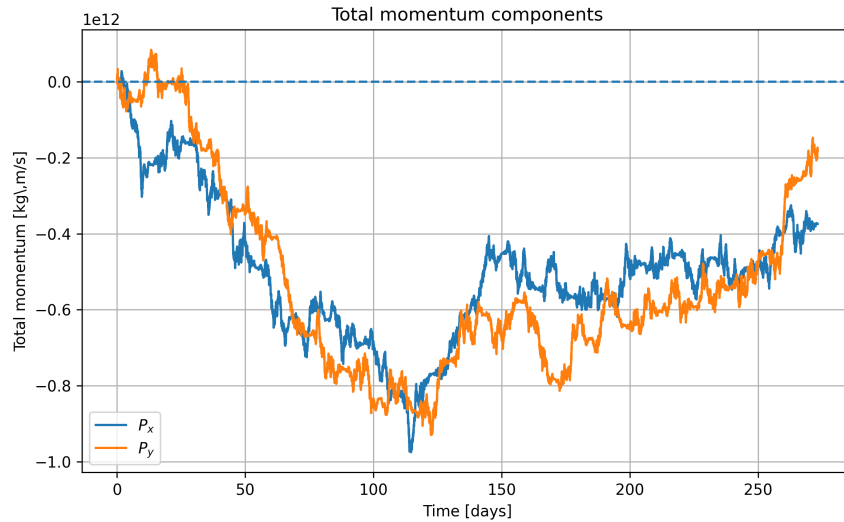


Figure 23: Momentum components over time

All the required physical properties are satisfied: the Earth–Moon distance remains constant, and both mechanical energy and total momentum are conserved up to very small numerical errors. These results validate the correctness and accuracy of the numerical simulation over long integration times.

3.4 Earth-Moon-Artemis II system

3.4.1 (a) Numerical solution for the return trajectory without atmosphere

In this part, we aim to determine the direction of the initial velocity of Artemis II such that its trajectory reaches a minimum altitude of

$$h_{\min} = 10\,000 \text{ m}$$

above the Earth's surface. The simulations are performed in the Earth-Moon center-of-mass frame, using the adaptive RK4 integration scheme and neglecting atmospheric effects ($\rho_0 = 0$).

The initial speed of the spacecraft is fixed to $|\mathbf{v}_0| = 1.2 \text{ km/s}$, and only the direction of the velocity is varied. For each value of the angle θ , the trajectory is simulated and the minimum altitude relative to the Earth's surface is computed as

$$h_{\min}(t) = r(t) - R_T,$$

where $r(t)$ is the distance between Artemis II and the center of the Earth, and R_T is the Earth's radius.

First of all, compared to Section 3.2, the initial conditions of Artemis II must be modified because the simulation is now performed in the Earth-Moon center of mass frame rather than in the Earth's frame. In Section 3.2, the position and velocity of the spacecraft were defined relative to the Earth, which was assumed to be at rest. In contrast, in the center of mass frame both the Earth and the Moon are in motion, and therefore Artemis II must be initialized consistently with this moving reference frame.

More precisely, the initial position of Artemis II is first defined relative to the Earth's position, and then expressed in the center of mass frame by adding the Earth's coordinates:

$$\mathbf{r}_A = \mathbf{r}_T + \mathbf{r}_{AT},$$

where $\mathbf{r}_{A/T}$ is the position of Artemis relative to the Earth. Similarly, the initial velocity of Artemis II is obtained by adding the Earth's velocity in the CM frame to the velocity defined in the Earth-centered frame:

$$\mathbf{v}_A = \mathbf{v}_T + \mathbf{v}_{AT}.$$

We have first done a broad scan of θ to find the relevant values and to focus our search. These values happen to lie around $\theta \approx 191^\circ$. This is consistent with the fact that Artemis II initially lies on the side of the Earth opposite to the Moon, and must therefore be directed approximately towards the Earth, 180 degrees. However, due to the motion of the Earth in the center of mass frame, the optimal direction is slightly shifted away from exactly 180° .

The results after the exhaustive search in this angular interval are shown in Figure 24. Where we can see the minimum altitude as a function of the initial angle θ in the refined region. The curve is approximately linear in this range, allowing for an accurate determination of the optimal angle.

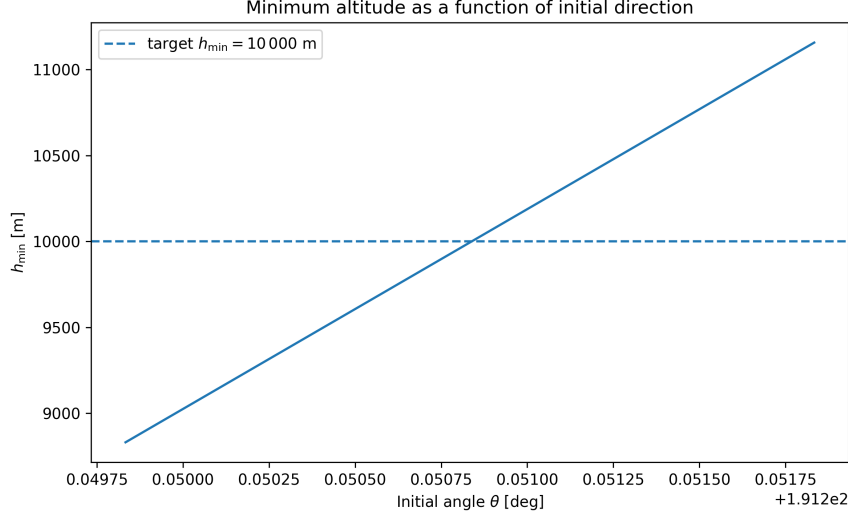


Figure 24: Minimum altitude of the Artemis II as a function of the initial velocity direction

The optimal angle we found is

$$\theta = 191.25083^\circ,$$

which yields a minimum altitude of

$$h_{\min} = 9994.14 \text{ m},$$

corresponding to an error of only

$$|h_{\min} - 10\,000| = 5.86 \text{ m}.$$

This satisfies the required precision of 100 m.

3.4.2 (b) Return trajectory with atmosphere

We performed a series of simulations using the adaptive time scheme, varying the direction of the initial velocity of Artemis II while keeping its magnitude fixed. For each trajectory, we computed the maximum acceleration experienced by the spacecraft, $a_{\max}$, and searched for the initial angle minimizing this quantity.

The results that can be seen in Fig. 25 show that the maximum acceleration decreases monotonically over the explored range of initial angles. This indicates that trajectories corresponding to larger angles lead to increasingly shallow atmospheric entries, and therefore to weaker aerodynamic drag. As a consequence, the minimum value of $a_{\max}$ within the explored domain is found at the boundary of the angular interval, specifically at $\theta = 191.31^\circ$.

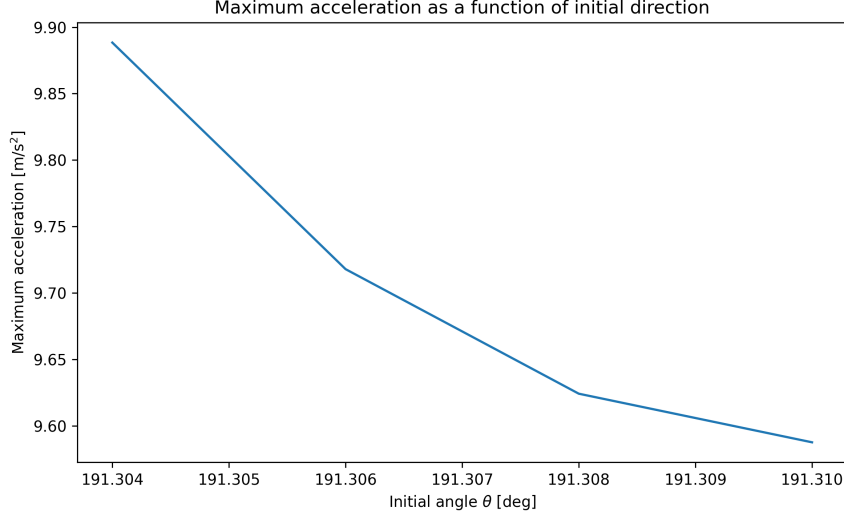


Figure 25: Maximum acceleration as a function of the initial velocity direction

In the adopted model, the atmospheric density follows an exponential profile of the form $\rho(h) = \rho_0 \exp(-h/H)$, which does not provide a sharp boundary between the atmosphere and vacuum. Therefore, we used a practical criterion to distinguish valid trajectories that effectively interact with the atmosphere. We considered a valid trajectory if it reaches a minimum altitude below $h_{\text{atm}} = 120$ km. Additionally, we impose that the spacecraft does not collide with the Earth, i.e. $h_{\text{min}} > 0$.

In order to verify whether a true minimum exists, we extended the angular search to a wider interval. The results confirm that the maximum acceleration continues to decrease as the initial angle increases, without exhibiting any internal minimum in the explored domain. This behavior is physically consistent: trajectories that enter the atmosphere more tangentially undergo weaker drag forces, leading to lower peak accelerations. As a result, the optimization problem does not admit a strict global minimum within the set of admissible trajectories, but rather favors increasingly shallow entries.

This result highlights a fundamental trade-off in atmospheric re-entry. While shallow trajectories minimize the mechanical stress experienced by the spacecraft, they also reduce the efficiency of energy dissipation. In practice, an optimal re-entry trajectory would need to balance these competing effects, which are not fully captured by the sole minimization of the maximum acceleration.

3.5 Earth-Moon and 3rd bodies, Lagrange points (Facultatifs)

3.5.1 Effective potential in the rotating frame

In this section we consider the motion of a third body of negligible mass in the rotating reference frame R' associated with the Earth–Moon system. This frame is centered at the center of mass G and rotates with angular velocity Ω such that the Earth and the Moon remain fixed on the x' -axis. Let (x', y') denote the position of the third body in this frame. The Earth and Moon are located at positions $(-a_T, 0)$ and $(a_L, 0)$ respectively, where as we saw in the previous section:

$$a_T = \frac{m_L}{m_T + m_L} d, \quad a_L = \frac{m_T}{m_T + m_L} d,$$

and d is the Earth–Moon distance.

The distances from the third body to the Earth and the Moon are therefore given by

$$r_T = \sqrt{(x' + a_T)^2 + y'^2}, \quad r_L = \sqrt{(x' - a_L)^2 + y'^2}.$$

The effective potential per unit mass in the rotating frame is obtained by combining the gravitational potentials of the Earth and the Moon with the potential associated with the centrifugal force

$$E'_{\text{pot}}(x', y') = -\frac{Gm_T}{r_T} - \frac{Gm_L}{r_L} - \frac{1}{2}\Omega^2(x'^2 + y'^2). \quad (7)$$

The first two terms correspond to the gravitational attraction of the Earth and the Moon, while the last term arises from the centrifugal force in the rotating frame. The Coriolis force does not contribute to the potential, as it cannot be derived from a scalar potential. We can see this potential in the following figure.

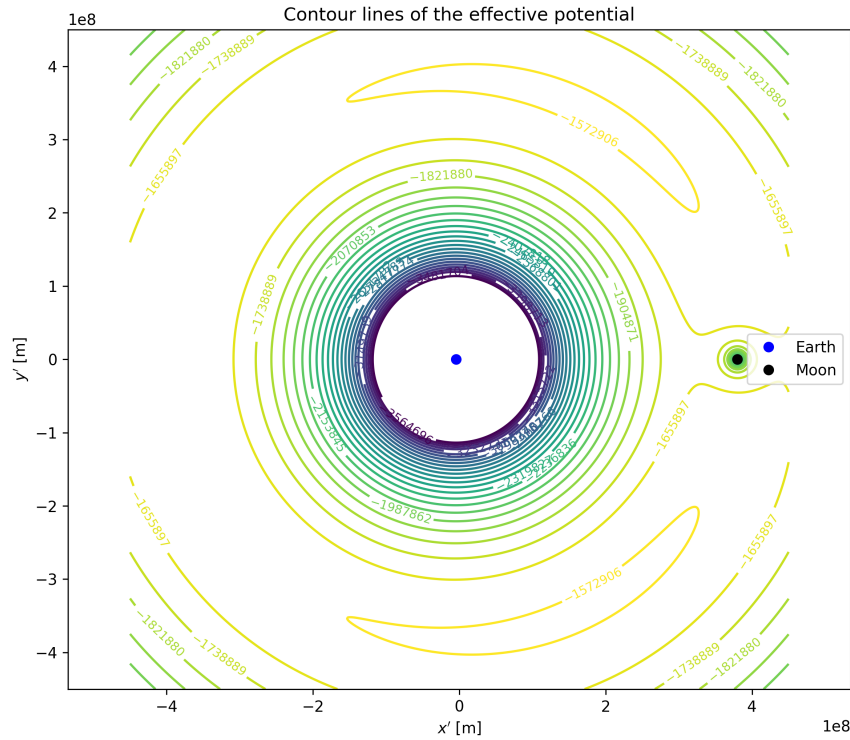


Figure 26: Contour lines of the effective potential

3.5.2 Lagrange point L4

For this section we place the third body at the Lagrange point L_4 , forming an equilateral triangle with the Earth and the Moon in the center of mass frame. Since the Earth and the Moon are located at $(-a_T, 0)$ and $(a_L, 0)$ respectively, the initial position of the third body is given by

$$x_3(0) = \frac{-a_T + a_L}{2}, \quad y_3(0) = \frac{\sqrt{3}}{2}d,$$

which corresponds to the vertex of an equilateral triangle of side length d .

In order for the third body to co-rotate with the Earth–Moon system, its initial velocity must match the rigid rotation of the system. In the inertial CM frame, this is given by

$$\vec{v}_3(0) = \vec{\Omega} \times \vec{r}_3,$$

which yields the components

$$v_{x,3}(0) = -\Omega y_3(0), \quad v_{y,3}(0) = \Omega x_3(0).$$

We have done a numerical simulation of the three-body system over a time span of one year. The trajectories in Fig. 27 show that the third body follows a circular orbit around the center of mass with the same angular velocity as the Earth Moon system. As a result, the triangular configuration is preserved throughout the simulation.

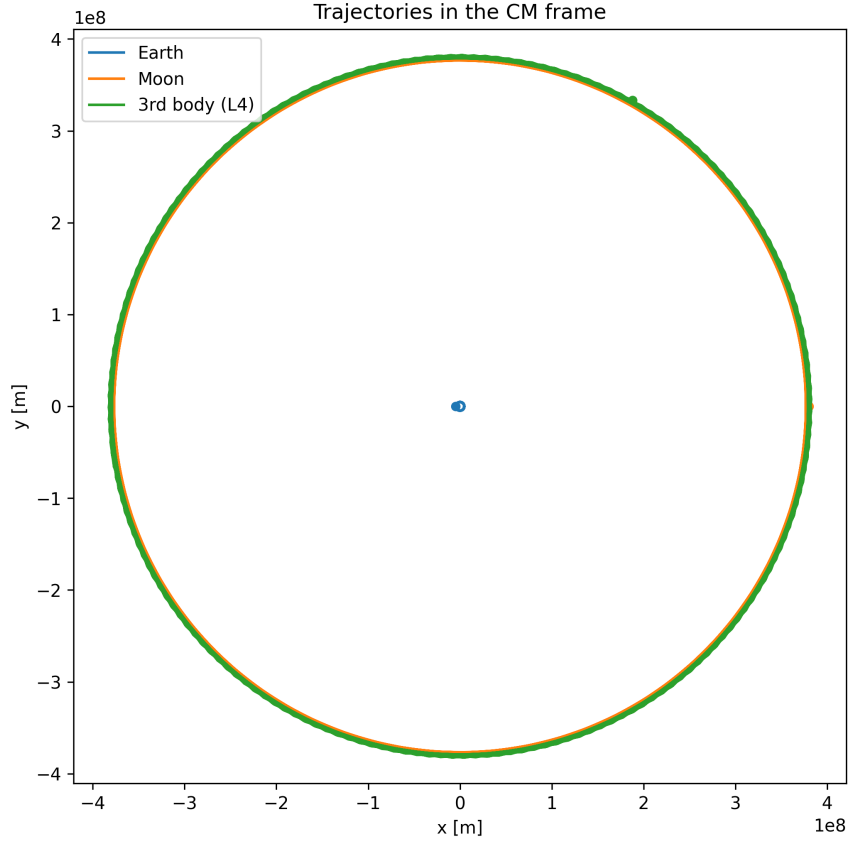


Figure 27: Trajectories in the center of mass frame

To quantitatively verify this behaviour, we computed the distances between the third body and the Earth, $d_{31}(t)$, and between the third body and the Moon, $d_{32}(t)$. The results in Fig. 28 show that both distances remain equal and nearly constant over time. In particular, the standard deviations obtained are of the order of 10^{-4} m, which is negligible compared to the characteristic scale of the system ($\sim 10^8$ m).

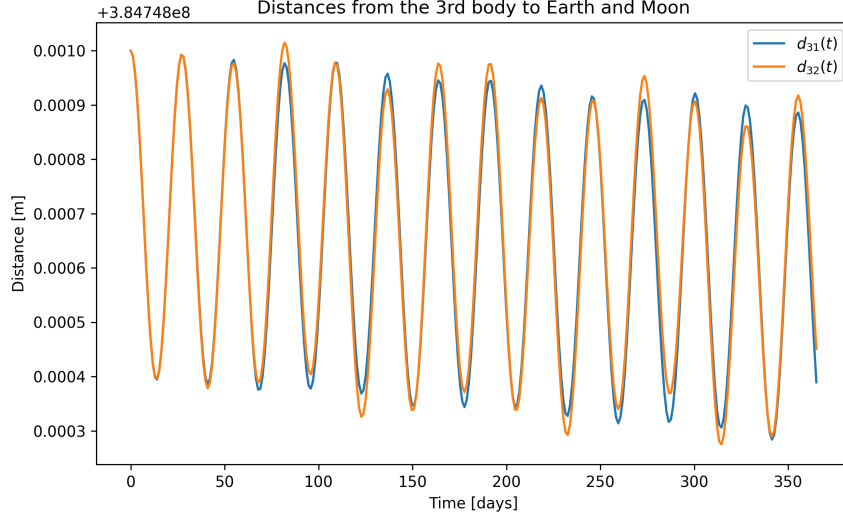


Figure 28: Distances from the 3rd body to Earth and Moon

This demonstrates that the third body remains at equal distance from the Earth and the Moon during the entire simulation, confirming numerically the stability of the Lagrange point L_4 in this model.

3.5.3 Displaced Lagrange point L_4

We have displaced the third body by 1000 km from the exact Lagrange point L_4 and represented the trajectories in the rotating Earth–Moon frame R' . In this frame, the Earth and the Moon remain fixed, while the third body no longer stays exactly at L_4 , since the equilibrium has been perturbed.

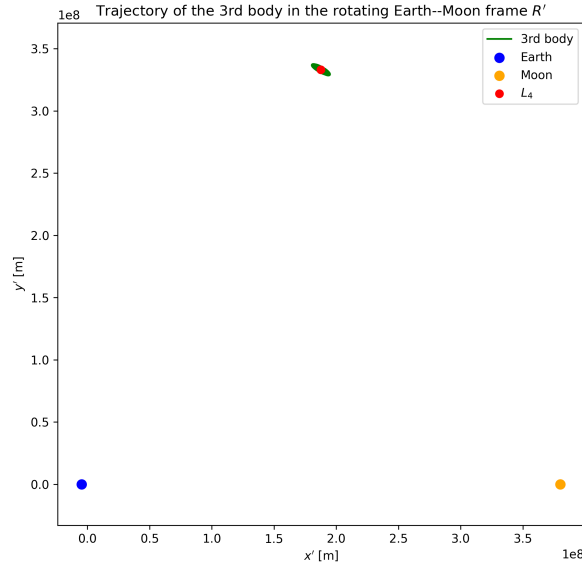


Figure 29: Trajectories of the 3rd body in the rotating R' frame

The numerical results show that the third body does not escape from the L_4 , but instead follows a bounded trajectory around it. This confirms that, for a small perturbation, the motion remains localized near the Lagrange point. The trajectory therefore illustrates the stability of L_4 in the Earth–Moon system within the framework of this model.

3.5.4 Other Lagrange points and stability

The Lagrange points are obtained as stationary points of the effective potential in the rotating Earth–Moon frame. Therefore, they are found by imposing

$$\nabla E'_{\text{pot}} = 0.$$

Using

$$E'_{\text{pot}}(x', y') = -\frac{Gm_T}{r_T} - \frac{Gm_L}{r_L} - \frac{1}{2}\Omega^2(x'^2 + y'^2),$$

with

$$r_T = \sqrt{(x' + a_T)^2 + y'^2}, \quad r_L = \sqrt{(x' - a_L)^2 + y'^2},$$

the two components of the gradient are

$$\begin{aligned} \frac{\partial E'_{\text{pot}}}{\partial x'} &= \frac{Gm_T(x' + a_T)}{r_T^3} + \frac{Gm_L(x' - a_L)}{r_L^3} - \Omega^2 x', \\ \frac{\partial E'_{\text{pot}}}{\partial y'} &= \frac{Gm_T y'}{r_T^3} + \frac{Gm_L y'}{r_L^3} - \Omega^2 y'. \end{aligned}$$

Hence, the Lagrange points satisfy

$$\frac{\partial E'_{\text{pot}}}{\partial x'} = 0, \quad \frac{\partial E'_{\text{pot}}}{\partial y'} = 0.$$

For the collinear points L_1 , L_2 , and L_3 , one has $y' = 0$, so the problem reduces to solving

$$\frac{Gm_T(x' + a_T)}{|x' + a_T|^3} + \frac{Gm_L(x' - a_L)}{|x' - a_L|^3} - \Omega^2 x' = 0$$

along the x' -axis. Numerically, we obtained

$$L_1 = (3.21994872 \times 10^8, 0),$$

$$L_2 = (4.44651520 \times 10^8, 0),$$

$$L_3 = (-3.86696397 \times 10^8, 0).$$

The L_5 point is obtained geometrically, since together with the Earth and the Moon they form an equilateral triangle of side length d . In particular,

$$L_5 = (1.87697755 \times 10^8, -3.33201542 \times 10^8).$$

In the next figures we have represented the motion of the perturbed third body in the rotating Earth Moon frame near the different Lagrange points.

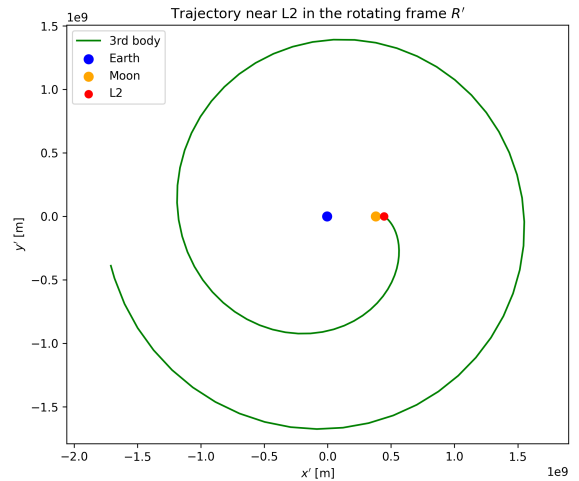
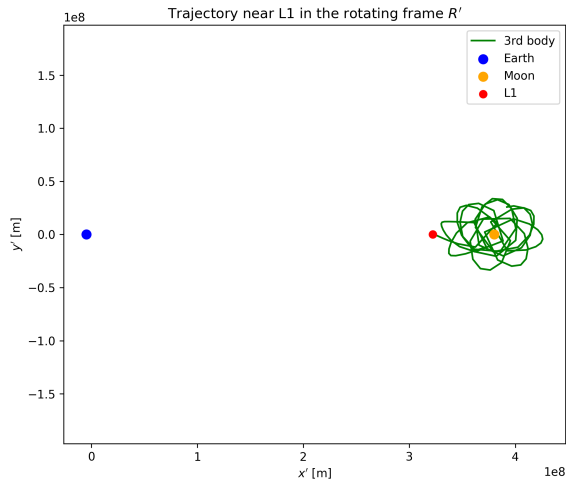


Figure 30: Trajectories of the 3rd body in L1

Figure 31: Trajectories of the 3rd body in L2

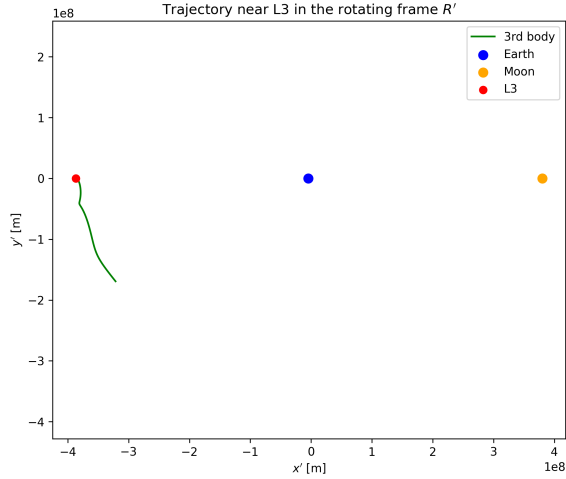


Figure 32: Trajectories of the 3rd body in L_3

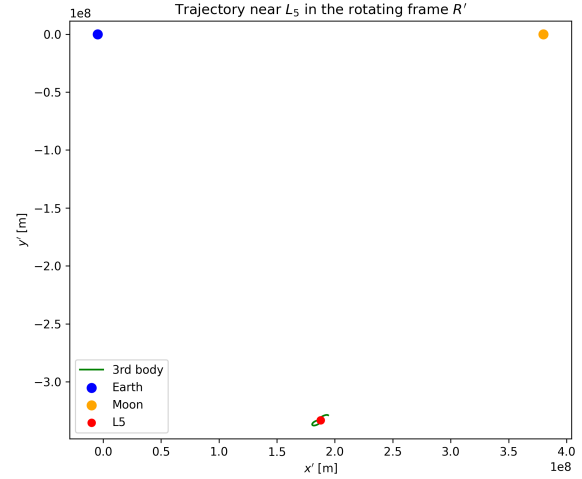


Figure 33: Trajectories of the 3rd body in L_5

Near L_1 , the trajectory remains temporarily in the neighbourhood of the equilibrium point but does not stay exactly localized there. This indicates that L_1 is not truly stable: a small perturbation is enough to produce a noticeable deviation from the equilibrium configuration.

The instability is even clearer for L_2 and L_3 . In both cases, the perturbed body rapidly drifts away from the corresponding Lagrange point, showing that these collinear equilibria are unstable. In particular, the trajectory near L_2 expands significantly, while near L_3 the body also leaves the vicinity of the equilibrium point instead of remaining bounded around it. By contrast, the trajectory near L_5 remains confined to a small region around the equilibrium point. This bounded motion indicates that L_5 is stable under small perturbations in the Earth–Moon system. This behaviour is consistent with the one previously observed for L_4 , confirming that the triangular Lagrange points L_4 and L_5 are stable, whereas the points L_1 , L_2 , and L_3 are unstable.

4 Conclusion

This project allowed us to validate the RK4 integration schemes, both with fixed and adaptive time steps, on gravitational systems of increasing complexity. The numerical results reproduce very accurately the analytical predictions for the Artemis return trajectory and for the Earth–Moon circular motion, while atmospheric drag highlights the importance of adaptive resolution during re-entry. Finally, the three-body simulations confirm the expected behaviour of the Lagrange points, with L_4 and L_5 stable under small perturbations, whereas L_1 , L_2 , and L_3 are unstable.

5 References

1. Exercise statement: *Physique Numérique – Exercice 3, Système gravitationnel à 1,2 et 3 corps. Artemis II, rentrée dans l’atmosphère.*
2. L. Villard, *Physique Numérique.*
3. Use of LLM tools (ChatGPT) for language correction and report rephrasing.